

Kit Line-Side Parts from Bins SOP (1x Episode: 6 Bins)

One episode builds one kit: parts are pulled from all six bins into one kit tray, every compartment is count-verified against the pick card, and the tray is tagged. The table begins with the **kit tray** on the **kit spot**, just right of the center of the table. The tray has six **compartments** in one row, numbered **1** at the left to **6** at the right. A **bin rack** stands directly behind the tray with six **bins**, numbered the same way, so **bin 1 sits behind compartment 1, bin 2 behind compartment 2**, and so on to bin 6. Each bin holds many of one part type. The **pick card** stands in a card stand at the back-left corner and says how many parts each bin gives. A **tag holder** holds upright **kit tags**.

The order never changes: fill compartments **1 through 6** from the bin behind each one; then count-verify compartments **1 through 6** against the pick card; then tag the tray. No compartment is counted until all six are filled. The tag does not go into the tray until every count matches the card. The episode does not end until the tag stands in the tag slot.

The right gripper reads the pick card, picks every part, places every part, spreads and counts every compartment, and puts the tag in. The left gripper presses the kit tray flat against the table by its left edge and holds it there for the whole episode, so the tray cannot slide while parts go in, while a compartment is spread for counting, or while the tag is pushed home. The left gripper stays on the left edge of the tray and never reaches across it. The right gripper works the tray, the bin rack, and the tag holder.

The table is set up in one of three ways. Only the tag holder moves; the kit tray, the bin rack, and the pick card are in the same place in all three.

- **Config M:** the tag holder is at the front-center, in front of the left end of the tray, clear of the left gripper.
- **Config R1:** the tag holder is at the front-right.
- **Config R2:** the tag holder is in the back-right corner, right of bin 6.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** one config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the left gripper is pinned on the tray's left edge for the whole episode, so no left zone is used and the **right gripper** takes the tag in every config. No arm reaches across the table.

- **Fixed roles:** everything else is the same in all three configs — the left gripper holds the tray, the right gripper reads, picks, places, fans, counts, and tags, and the fill, verify, tag order does not change.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the kit tray with all six compartments, the bin rack with all six bins, the pick card in its stand at the back-left, and the tag holder in the start zone for this episode's config.
3. The kit tray is visible from above, so the inside of all six compartments, the compartment numbers, and the tag slot on the front edge can all be seen.
4. Both arms are at home with grippers open.
5. The table is bare apart from the kit tray, the bin rack, the pick card stand, the tag holder, and robot hardware.
6. The right arm reaches bin 1, bin 6, compartment 1, compartment 6, and the tag holder — front-center (Config M), front-right (Config R1), or back-right (Config R2) — without stretching. The left arm reaches the left edge of the kit tray without stretching.

Materials checklist

1. One **kit tray** sits on the **kit spot**, just right of the center of the table, **square** to the front edge and flat on the table.
2. The tray has six **compartments** in one row, split by **dividers**, numbered **1** at the left to **6** at the right. The numbers are printed on the front rail of the tray and are readable from above.
3. Every compartment starts **empty**.
4. The **tag slot** is on the front edge of the tray, in the middle, and is empty.
5. The **bin rack** stands directly behind the tray, over the same span, with six open-front **bins** numbered **1** at the left to **6** at the right. **Bin 1 sits behind compartment 1**, bin 2 behind compartment 2, and so on to bin 6.
6. Each bin holds many of **one part type** only, and holds more parts than the pick card asks for: bin 1 hex bolts, bin 2 flat washers, bin 3 lock washers, bin 4 hex nuts, bin 5 nylon spacers, bin 6 spring clips.

7. The **pick card** stands upright in a card stand at the **back-left corner**, angled toward the front, and reads: bin 1 — **2** parts, bin 2 — **3** parts, bin 3 — **1** part, bin 4 — **2** parts, bin 5 — **3** parts, bin 6 — **1** part. **Twelve parts in all.**
8. The **tag holder** holds six identical **kit tags** standing upright, printing facing the front, and stands on the **tag zone** for this episode's config:
 - **Config M:** front-center, in front of the left end of the tray, clear of the left gripper
 - **Config R1:** front-right
 - **Config R2:** back-right corner, right of bin 6
9. Keep the left side of the table clear. The left gripper comes in from there onto the tray's left edge.
10. Before collection, confirm by hand that:
 - one part can be pinched out of each bin without dragging others with it;
 - a part laid in a compartment stays inside it and does not bounce out;
 - the parts a bin gives fit in one compartment in a **single layer**;
 - a kit tag slides into the tag slot and stands there on its own.

Workspace layout

- **Kit spot:** just right of the center of the table — the kit tray stays here all episode
- **Kit tray:** compartments 1 to 6, left to right, with the tag slot on the front edge
- **Behind the tray:** the bin rack, bins 1 to 6, each bin directly behind its own compartment
- **Back-left corner:** the pick card in its stand
- **Tag zone:** the tag holder — front-center (Config M), front-right (Config R1), or back-right (Config R2)
- **Left side:** kept clear, so the left gripper can come in onto the tray's left edge

Arm assignments

- **Left gripper:** presses the kit tray flat by its left edge and holds it there for the whole episode, so the tray cannot slide while the right gripper works. It never takes a tag, in any config.
- **Right gripper:** reads the pick card, picks parts from the bins, places them in the compartments, spreads and counts each compartment, returns extras, and puts the kit tag in the tag slot, taking it from the tag holder wherever this episode's config puts it.

Vocabulary

- **Kit spot:** the place on the table, just right of center, where the kit tray sits. The tray stays here for the whole episode.

- **Square:** the front edge of the tray lines up with the front edge of the table, so the row of compartments runs level and neither end of the tray sits nearer the front.
- **Kit tray:** the tray the parts are kitted into. It stays on the kit spot for the whole episode and is never lifted.
- **Compartment:** one walled pocket in the kit tray. Each one has its own number, 1 at the left to 6 at the right.
- **Divider:** the wall between two compartments.
- **Compartment number:** the number printed on the front rail of the tray beside each compartment.
- **Bin rack:** the rack behind the tray that holds the six bins.
- **Bin:** one open-front box on the rack. Each bin holds one part type only and has its own number.
- **Bin number:** the number on the front of a bin, 1 at the left to 6 at the right. Bin 1 sits behind compartment 1, and so on to bin 6.
- **Part:** one item out of a bin — one bolt, one washer, one nut, one spacer, one clip.
- **Line-side parts:** the parts kept in the bins at the rack, ready to be kitted.
- **Pick card:** the card at the back-left that says how many parts each bin gives.
- **Quantity:** the number of parts the pick card asks for from one bin.
- **Pick:** take one part out of a bin with the right gripper.
- **Place:** set one part down inside a compartment with the right gripper and open the gripper there.
- **Single layer:** every part in the compartment sits on the tray floor, and no part sits on top of another.
- **Fanned flat:** the parts in a compartment have been spread out into a **single layer**, so each one can be seen and counted on its own.
- **Count-verify:** fan a compartment flat, count the parts in it, and compare that number with the quantity the pick card gives for that bin.
- **Short:** the compartment holds fewer parts than the card asks for.
- **Over:** the compartment holds more parts than the card asks for.
- **Straddling:** a part rests on a **divider**, or lies half in one compartment and half in the next.
- **Kit tag:** one of the printed cards in the tag holder. One tag goes into the tray each episode.
- **Tag zone:** where the tag holder stands for the episode — front-center (Config M), front-right (Config R1), or back-right (Config R2).
- **Tag holder:** the holder the kit tags stand in, on the tag zone.
- **Tag slot:** the slot on the front edge of the kit tray that the kit tag goes into.
- **Seated:** the kit tag is pushed down into the tag slot, stands upright with its printing facing the front, and stays there when the right gripper lets go.

Steps

Run Steps 1–4 in order on the one kit tray, then end the episode with Step 5. Only Step 4 depends on the config: the **right gripper** takes the tag from the tag holder at the front-center (Config M), front-right (Config R1), or back-right (Config R2) and carries it round the compartments to the tag slot. Every other line is the same in all three configs.

Step 1: Hold the kit tray down

Goal: the kit tray is pinned flat on the kit spot and cannot slide for the rest of the episode.

- With the **left gripper**, press down on the **left edge** of the kit tray and hold it against the table.
- Keep the **left gripper** there through Steps 2, 3, and 4.

Check: the tray is flat on the table, **square** to the front edge, and does not slide when the left gripper presses. If the tray is crooked, straighten it with the **left gripper** first, then press it down.

Expected state: the tray is held, all six compartments are empty, and the tag slot is empty.

Step 2: Fill all six compartments from the bins

Goal: every compartment holds the number of parts the pick card asks for, taken from the bin behind it.

- With the **left gripper**, keep pressing the kit tray down.
- Work **bin 1 first, then 2, 3, 4, 5, 6**. Finish one bin before starting the next.
- With the **right gripper**, move **one part at a time**. Never scoop, never carry two parts, and never tip a bin out.
- Carry each part straight forward, from its bin to the compartment right in front of it. The **right gripper** never carries a part over another compartment.
- Keep every carry low over the table. Do not lift a part high above the tray.

2.1 Read the quantity

- With the **right gripper** still clear of the tray, read the **pick card** for the bin being worked and note its **quantity**.
- Read the card again for every bin. Do not work from memory.

2.2 Pick and place one part

- With the **right gripper**, reach into that bin and pinch **one part**.
- Lift it straight up and clear of the bin lip, then carry it straight forward to the compartment with the same number.
- Lower the part into the compartment until it is inside the walls, then open the **right gripper** and let go.

- Do not drop or toss a part in from above the tray.
- If more than one part comes out together, put the extra parts back in that same bin with the **right gripper** before going on.
- Repeat 2.2 until that bin's **quantity** has been placed, then go back to 2.1 for the next bin.

Check: after each bin, the number of parts in the compartment matches the card, every part lies inside its own compartment, and no part is **straddling** a divider or sitting in a neighbor. If a part is straddling or in the wrong compartment, pick it up with the **right gripper** and place it in the right one. If a part is on the table, pick it up with the **right gripper** and put it back in its own bin.

Expected state: all six compartments hold parts, each part type matches its bin, the bins are still on the rack, the pick card is still standing in its stand, and the tag slot is still empty.

Step 3: Count-verify all six compartments

Goal: every compartment has been fanned flat, counted, and matched against the pick card.

- With the **left gripper**, keep pressing the kit tray down.
- Count-verify **compartment 1 first, then 2, 3, 4, 5, 6**. Verify every compartment, every episode.
- Do not put the kit tag in until all six compartments have been verified.

3.1 Fan the compartment flat

- Close the **right gripper** and bring its tips down into the compartment.
- Sweep the tips gently across the floor of the compartment to spread the parts into a **single layer**, so no part sits on another.
- Lift the **right gripper** clear of the tray so nothing is hidden under it.

3.2 Count against the card

- Count the parts now visible in that compartment.
- Read the **pick card** for that bin number and compare the two numbers.
- If the count matches, move the **right gripper** on to the next compartment.
- If the compartment is **short**, pick the missing parts one at a time from that same bin with the **right gripper**, place them, then fan and count that compartment again.
- If the compartment is **over**, pick the extra parts one at a time with the **right gripper** and put them back in **that compartment's own bin**, then fan and count that compartment again.

Check: all six compartments were fanned flat and counted, and each count matches the pick card. If any compartment was not counted, fan and count it with the **right gripper** before going on to Step 4.

Expected state: all six compartments hold the quantity the card asks for, every compartment is in a single layer, no part is straddling a divider, and the bins are still on the rack.

Step 4: Tag the tray

Goal: one kit tag stands seated in the tag slot on the front edge of the tray.

- With the **left gripper**, keep pressing the kit tray down.
- **IF the tag holder is at the front-center (Config M):** with the **right gripper**, pinch **one kit tag** by its top edge, lift it straight up out of the **tag holder**, and carry it right along the front edge of the tray to the **tag slot**.
- **IF the tag holder is at the front-right (Config R1):** with the **right gripper**, pinch **one kit tag** by its top edge, lift it straight up out of the **tag holder**, and carry it left along the front edge of the tray to the **tag slot**.
- **IF the tag holder is at the back-right (Config R2):** with the **right gripper**, pinch **one kit tag** by its top edge, lift it straight up out of the **tag holder**, and carry it forward past the right end of the tray, then left along the front edge to the **tag slot**.
- Then, in all three:
- Carry the tag in one go. Do not set it down on the way, and do not carry it over the open compartments.
- With the **right gripper**, line the tag up with the slot, printing facing the front, and push it straight down into the slot until it stands on its own.
- Open the **right gripper** and let go. The tag must stay upright.

Check: one tag is **seated** — upright in the slot, printing facing the front, and it stays there when the right gripper lets go. If the tag leans, falls, or faces backwards, take it out with the **right gripper**, turn it printing-forward, and push it down again.

Expected state: the tray holds twelve parts across six compartments, every count matches the card, and one tag stands seated in the tag slot.

Step 5: End the episode

Goal: recording ends with the kit filled, verified, and tagged.

1. Confirm the end state:

- all six compartments hold the quantity the pick card asks for;
- each compartment holds only the part type from the bin behind it;
- no part is **straddling** a divider and no part is loose on the table;
- one kit tag stands **seated** in the tag slot, printing facing the front;
- all six bins are still on the rack and the pick card is still standing in its stand;
- the kit tray is still **square** on the kit spot.

2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

After each episode

1. Pull the kit tag out of the tag slot and stand it back in the tag holder, printing facing the front. Set the tag holder on the tag zone for the next episode's config — front-center (Config M), front-right (Config R1), or back-right (Config R2).
2. Empty each compartment back into its own bin, compartment 1 into bin 1 through compartment 6 into bin 6. Do not mix part types between bins.
3. Pick up any part that ended up on the table or the floor and put it back in its own bin.
4. Brush dust and chips out of the six compartments and the tag slot.
5. Set the kit tray **square** on the kit spot, flat against the table, with all six compartments empty. Line the bin rack up behind it so bin 1 sits behind compartment 1 and bin 6 behind compartment 6.
6. Stand the pick card upright in its stand at the back-left, readable from the front.
7. Run both Setup checklists before the next episode.

At the end of the session

1. Inspect the parts and the fixtures. Throw out any part that is bent, burred, or broken. Top each bin back up so it holds more parts than the pick card asks for. Replace the tray if a divider is cracked or the tag slot no longer holds a tag upright. Replace a kit tag that is bent, torn, or soft. Replace the pick card if it is creased or no longer readable.
2. Leave the tray square on the kit spot with all six compartments empty, the six bins full on the rack in number order, six tags standing in the tag holder, and the pick card upright in its stand.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config R1 (tag holder at the front-right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the tag holder is not on the tag zone for the config; a gripper reaches across the table for the tag; the left gripper leaves the tray to take the tag; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (one config per episode, chosen before recording; the right gripper takes the tag in every config and no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the tag holder is before the tag reach, then follow that config's IF line in Step 4.

Violation: Kit tray not held

- **Visible cue:** the left gripper is off the left edge of the kit tray while the right gripper places a part, fans a compartment, or pushes the tag in, and the tray slides, lifts, or is chased across the spot.
- **SOP rule broken:** Step 1 (the left gripper presses the kit tray down by its left edge and holds it there through Steps 2, 3, and 4).
- **Coaching note:** left gripper on the tray first, and leave it there until the episode ends.

Violation: Kit tray moved off the kit spot

- **Visible cue:** either gripper lifts, turns, or drags the kit tray, or the tray ends the episode out of square with the front edge.
- **SOP rule broken:** Steps 1–4 (the kit tray stays flat and square on the kit spot for the whole episode and is never lifted).
- **Coaching note:** bring the parts and the tag to the tray; never move the tray to them.

Violation: Bins worked out of order, or a bin skipped

- **Visible cue:** the right gripper works the bins in some order other than 1, 2, 3, 4, 5, 6, jumps back to a bin it already finished, or never picks from one of the six bins.
- **SOP rule broken:** Step 2 (work bin 1 first, then 2, 3, 4, 5, 6, finishing one bin before starting the next).
- **Coaching note:** left to right, one bin at a time. Say the bin number before you reach in.

Violation: Part taken from the wrong bin

- **Visible cue:** the right gripper reaches into one bin and places the part in a compartment with a different number, so a compartment ends up holding a part type from another bin.
- **SOP rule broken:** Step 2 (carry each part straight forward from its bin to the compartment right in front of it).
- **Coaching note:** straight forward, never sideways. Bin 4 feeds compartment 4 and nothing else.

Violation: Part placed in the wrong compartment or left straddling

- **Visible cue:** at the end of a bin, a part sits in a neighbor compartment, rests on a divider, or lies half in one compartment and half in the next, and the episode moves on.
- **SOP rule broken:** Step 2 (every part lies inside its own compartment; a part straddling a divider or in a neighbor is picked up with the right gripper and placed in the right one).
- **Coaching note:** look into the compartment after each part. Fix a straddler the moment you see it.

Violation: More than one part moved at a time

- **Visible cue:** the right gripper scoops, rakes, or lifts two or more parts together, tips a bin out, or carries a handful to a compartment, and the extras are not put back.
- **SOP rule broken:** Step 2 (move one part at a time; if more than one comes out together, put the extras back in that same bin before going on).
- **Coaching note:** one pinch, one part. If two come up, drop the spare back in the bin first.

Violation: Part carried over other compartments or high above the tray

- **Visible cue:** the right gripper travels across the open compartments on its way from a bin, or lifts a part high above the tray instead of a low straight-forward carry.
- **SOP rule broken:** Step 2 (carry each part straight forward from its bin to its own compartment, low over the table, never over another compartment).
- **Coaching note:** short, low, straight forward. A part dropped over the tray lands in the wrong pocket.

Violation: Part dropped in instead of placed

- **Visible cue:** the right gripper opens above the tray and the part falls, tosses, or bounces into the compartment rather than being lowered inside the walls and released there.
- **SOP rule broken:** Step 2 (lower the part into the compartment until it is inside the walls, then open the right gripper; do not drop or toss a part in from above).
- **Coaching note:** take it all the way down into the pocket before you open the gripper.

Violation: Wrong quantity taken from a bin

- **Visible cue:** a compartment ends Step 2 holding fewer or more parts than the pick card gives for that bin, or is left empty, and the right gripper moves on to the next bin.
- **SOP rule broken:** Step 2 (place that bin's quantity, no more and no fewer, before going on to the next bin).
- **Coaching note:** count out the number before you start the bin, and count what is in the pocket before you leave it.

Violation: Pick card not read, moved, or covered

- **Visible cue:** either gripper moves, knocks, or lays the card flat, something is set on top of it, or the right gripper works a bin and a compartment count that does not match the card without the card ever coming back into view.
- **SOP rule broken:** Steps 2 and 3 (read the pick card for every bin and every count; do not work from memory, and leave the card standing in its stand).
- **Coaching note:** the card is the count, not your memory. Read it, and leave it standing where it is.

Violation: Bin lifted, dragged, or spilled

- **Visible cue:** either gripper lifts a bin, drags it along the rack, tips it to pour parts out, or leaves parts spilled on the rack or the table.
- **SOP rule broken:** Step 2 (never tip a bin out; the bins stay on the rack in number order for the whole episode).
- **Coaching note:** reach into the bin, never move the bin. Pinch parts out one by one.

Violation: Count-verify skipped

- **Visible cue:** the right gripper fans and counts fewer than six compartments, or none at all, before the episode moves on to the tag or the ending.
- **SOP rule broken:** Step 3 (count-verify compartment 1 through compartment 6; verify every compartment, every episode).
- **Coaching note:** six compartments, six counts. Count them out loud as you go.

Violation: Parts not fanned flat before counting

- **Visible cue:** the right gripper counts a compartment while parts sit stacked on each other, or it never brings its tips into the compartment to spread them, or it stays over the compartment so parts are hidden under the gripper while counting.
- **SOP rule broken:** Step 3 (spread the parts into a single layer with the closed right gripper, then lift the gripper clear of the tray before counting).
- **Coaching note:** spread it out, pull the gripper back, then count what you can see.

Violation: Miscount found but not fixed

- **Visible cue:** a compartment counts short or over against the card and the right gripper moves on to the next compartment, or it adds or removes parts and never fans and counts that compartment again.
- **SOP rule broken:** Step 3 (on a short or over compartment, correct it from that bin and then fan and count that compartment again).
- **Coaching note:** every fix earns another count on that same pocket.

Violation: Extra parts not returned to their own bin

- **Visible cue:** the right gripper takes extras out of an over compartment and puts them in a different bin, into another compartment, or leaves them on the table or the rack.
- **SOP rule broken:** Step 3 (put extra parts back in that compartment's own bin).
- **Coaching note:** an extra goes home to the bin it came from, one part at a time.

Violation: Work done out of order

- **Visible cue:** the phases run out of sequence — the right gripper fans and counts a compartment before all six are filled, takes a tag before all six counts are verified, or goes back to fill a compartment after the tag is in.
- **SOP rule broken:** Steps 2–4 (fill all six, verify all six, then tag, in that order).
- **Coaching note:** finish the phase before you start the next one. Fill, count, tag.

Violation: Tray left untagged

- **Visible cue:** the episode ends with the tag slot empty, or with a tag lying loose in the tray or on the table instead of in the slot.
- **SOP rule broken:** Step 4 (one kit tag is pushed down into the tag slot before the episode ends).
- **Coaching note:** no kit leaves without its tag. The slot is the last thing you fill.

Violation: Tag not seated

- **Visible cue:** the tag leans, falls over, sits on top of the slot, or stands with its printing facing away from the front, and the right gripper leaves it that way.

- **SOP rule broken:** Step 4 (push the tag straight down into the slot, printing facing the front, and let go only once it stands on its own).
- **Coaching note:** printing to the front, push it down, then let go and watch that it stays.

Violation: Tag mishandled on the way

- **Visible cue:** the right gripper pulls more than one tag out of the holder, bends or creases a tag, sets a tag down on the way to the slot, or carries it over the open compartments.
- **SOP rule broken:** Step 4 (pinch one tag by its top edge and carry it in one go to the slot, not over the compartments and not set down on the way).
- **Coaching note:** one tag, top edge, straight to the slot.

Violation: Tag holder or bin rack knocked out of place

- **Visible cue:** either gripper tips, pushes, or drags the tag holder or the bin rack off its spot, spills the tags, or leaves the rack out of line with the tray so a bin no longer sits behind its own compartment.
- **SOP rule broken:** Steps 2 and 4 (the bins stay on the rack behind their own compartments and the tag holder stays on its spot at the front-right through the whole episode).
- **Coaching note:** reach in over the lip. Work slower and lower so nothing gets shoved.

Violation: Part or tag dropped and left

- **Visible cue:** either gripper drops a part or a tag onto the rack, the table, or the floor, and the episode moves on without it being picked up and put back.
- **SOP rule broken:** Step 2 (a part that ends up on the table is picked up with the right gripper and put back in its own bin) and Step 4 (a dropped tag is retrieved before the tray is tagged).
- **Coaching note:** pick up what you drop before the next pick. Nothing stays loose on the table.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper picking a part, placing a part, fanning a compartment, or handling a tag, or the right gripper holding down the tray.
- **SOP rule broken:** Steps 1–4 (the right gripper picks, places, fans, counts, and tags; the left gripper holds the tray).
- **Coaching note:** right gripper does the work, left gripper holds the tray. It does not swap.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the six counts and the tag are confirmed, an arm is not home, a gripper is closed, or an arm does something else after homing.

- **SOP rule broken:** Step 5 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A bin arrives holding fewer parts than the pick card asks for, so the compartment cannot be filled.
- A bin arrives holding two part types mixed together.
- A part is bent, burred, or broken, so it will not sit flat in the compartment.
- A tray divider is cracked or missing, so parts pass between compartments on their own.
- The tag slot is broken or spread open, so a correctly pushed tag will not stand.
- The pick card is misprinted or unreadable, so the quantities cannot be read.

Annotation subtasks (from SOP)

1. Press the kit tray flat and hold it with the left gripper
2. Read the pick card for the bin being worked
3. Pick one part from the bin with the right gripper
4. Place the part in the compartment in front of that bin
5. Repeat the pick and place until that bin's quantity is in
6. Repeat for all six bins, bin 1 through bin 6
7. Fan a compartment flat with the closed right gripper
8. Count the compartment against the pick card
9. Add missing parts or return extras to that bin, then count again
10. Repeat the fan and count for all six compartments
11. Take one kit tag from the tag holder
12. Push the tag down into the tag slot, printing facing the front
13. Confirm the end state, return both arms home, and end the episode